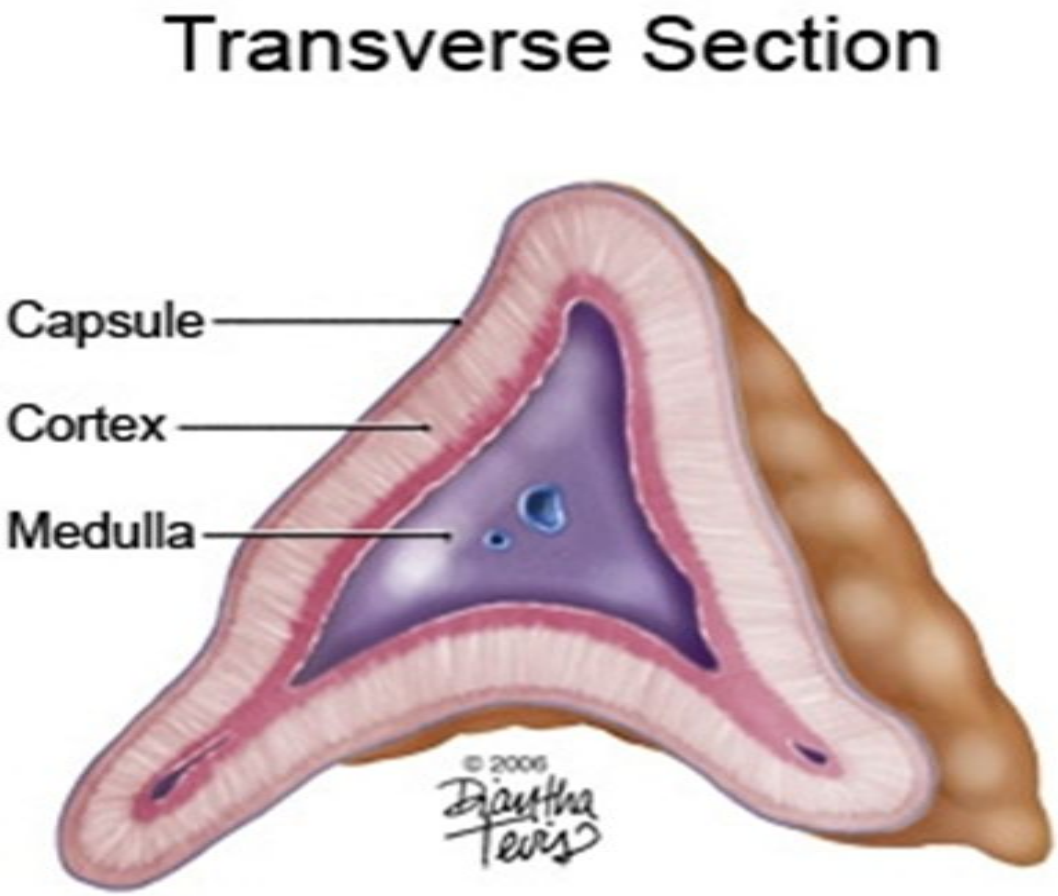
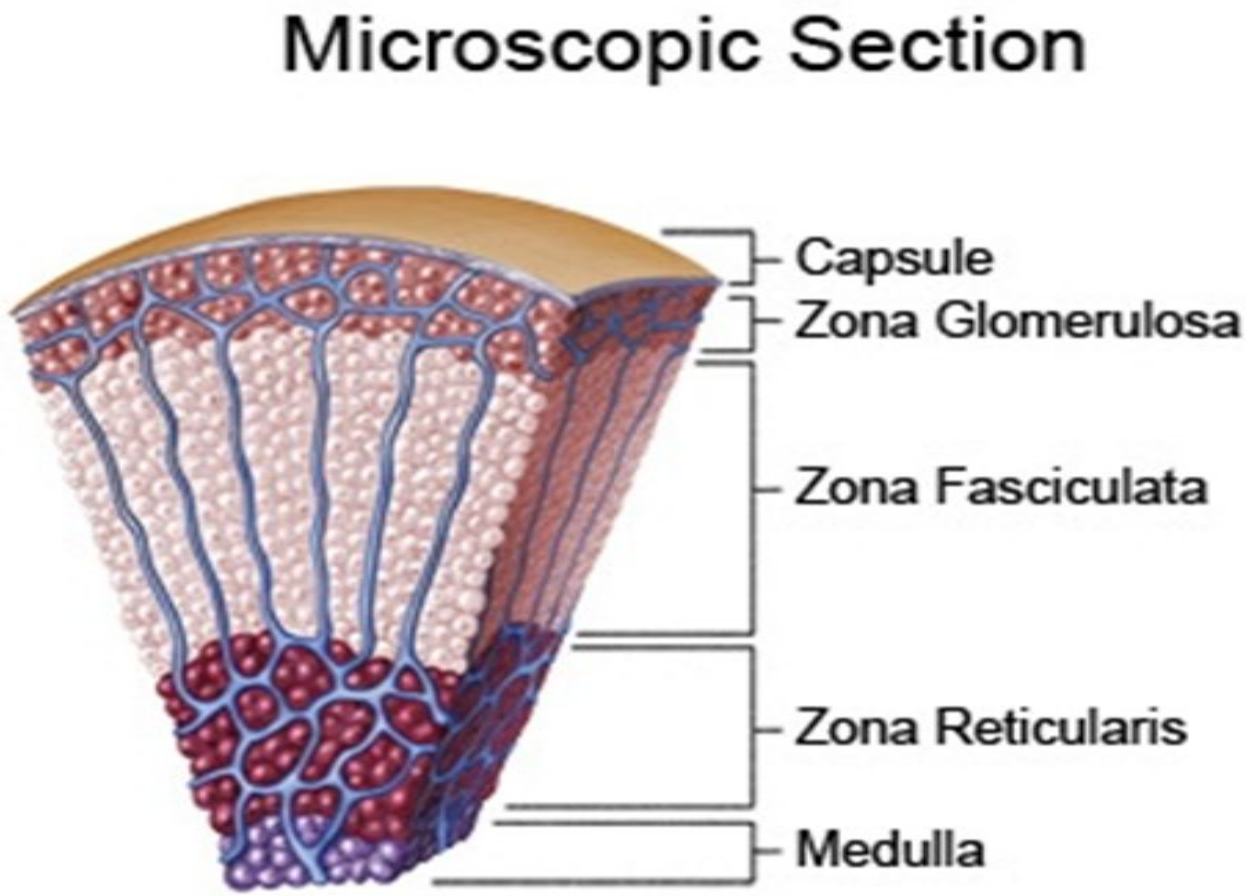
Adrenal gland: Structure, Hormones and clinical disorders





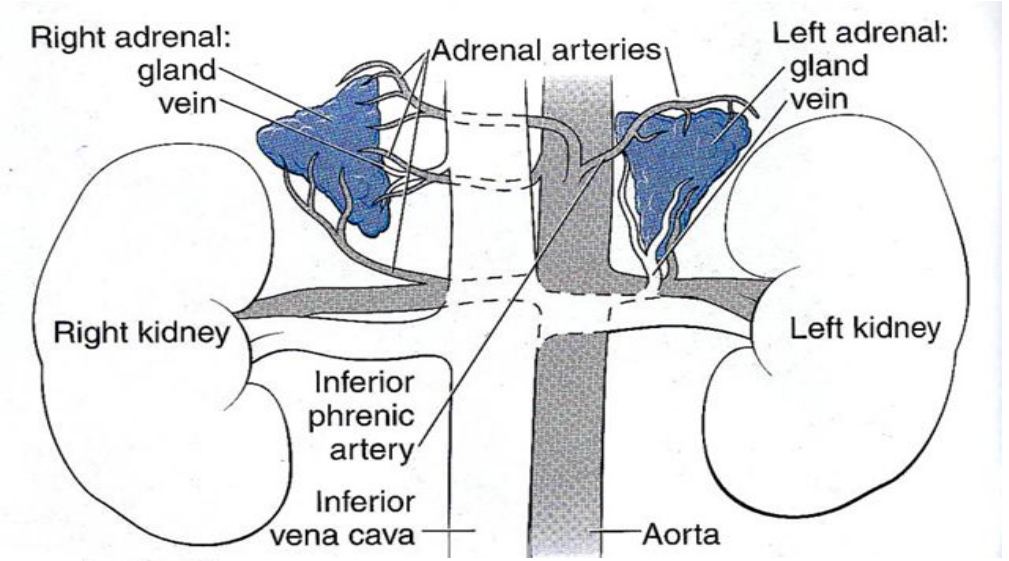
DR ABDUL SAMAD KHAN
SENIOR LECTURER DIKIOHS,
DOW UNIVERSITY OF HEALTH SCIENCES

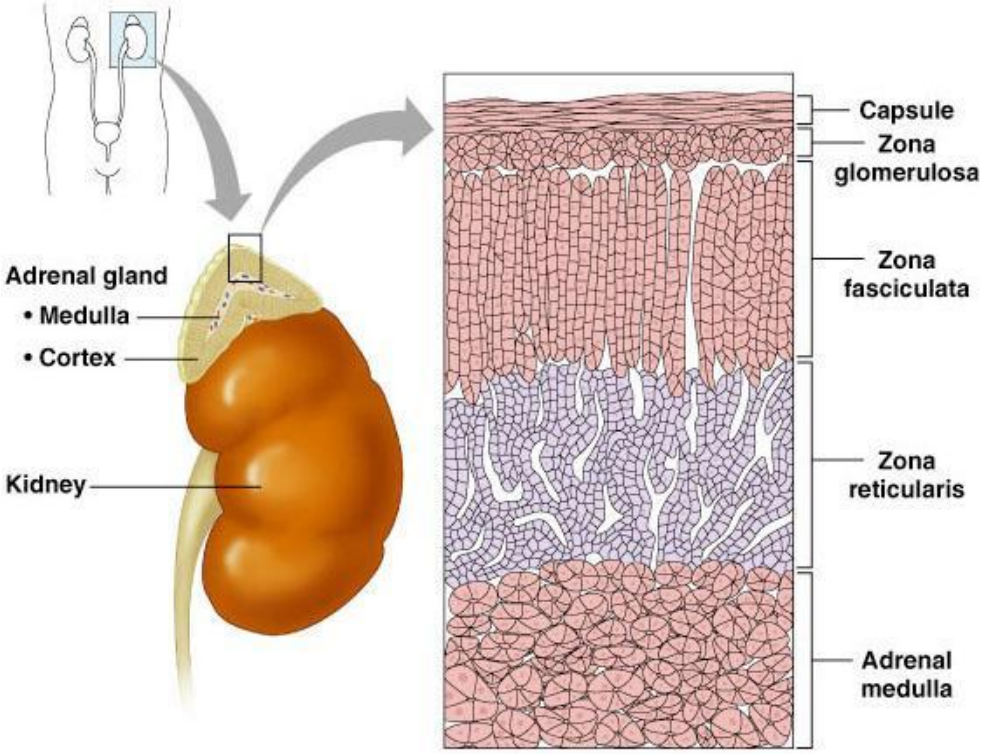
Anatomy of adrenal glands

Localization: At the top of the kidney and weighing approximately 5 g each.

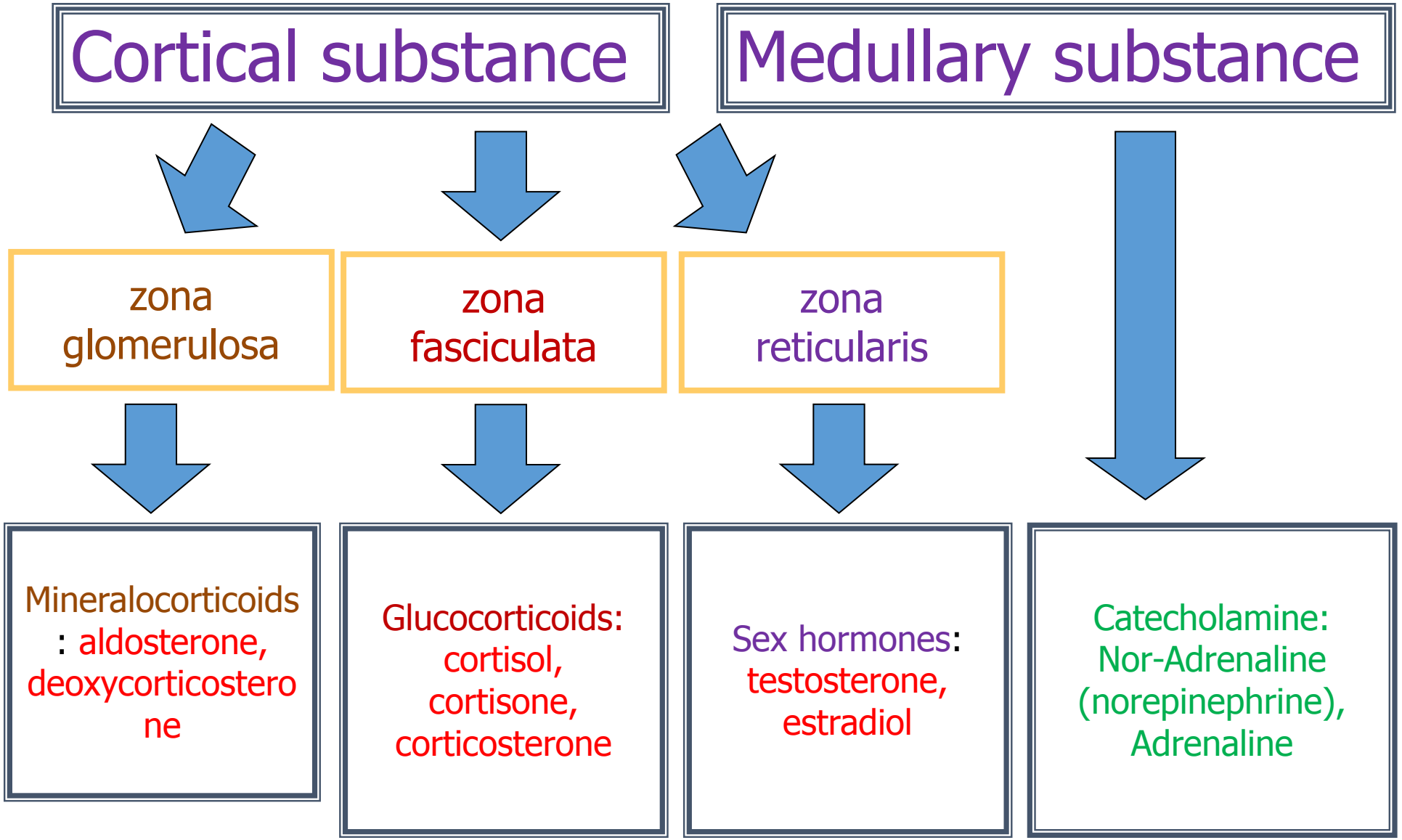
Each adrenal gland is composed of two endocrine components:

- ***Cortex*** (outer part) that constitutes the 80% of the gland. The cortex consists of three zones.
- **Medulla** (inner part) that constitutes 20% of the gland.
- The medulla and each of the zones in the cortex, each produce different hormones that serve a variety of functions in your body.





Adrenal glands



Functions of the The Adrenal Glands

Adrenal glands are called the ‘life-saving glands’ or ‘essential endocrine glands’.

It is because the absence of adrenocortical hormones causes death within 3 to 15 days and absence of adrenomedullary hormones, drastically decreases the resistance to mental and physical stress.

Maintaining homeostasis; Role in the regulation:

- of the body’s adaptive response to stress
- in the maintenance of body water, Na and K balance
- in the control of blood pressure

The main hormones

Adrenal Cortex:-

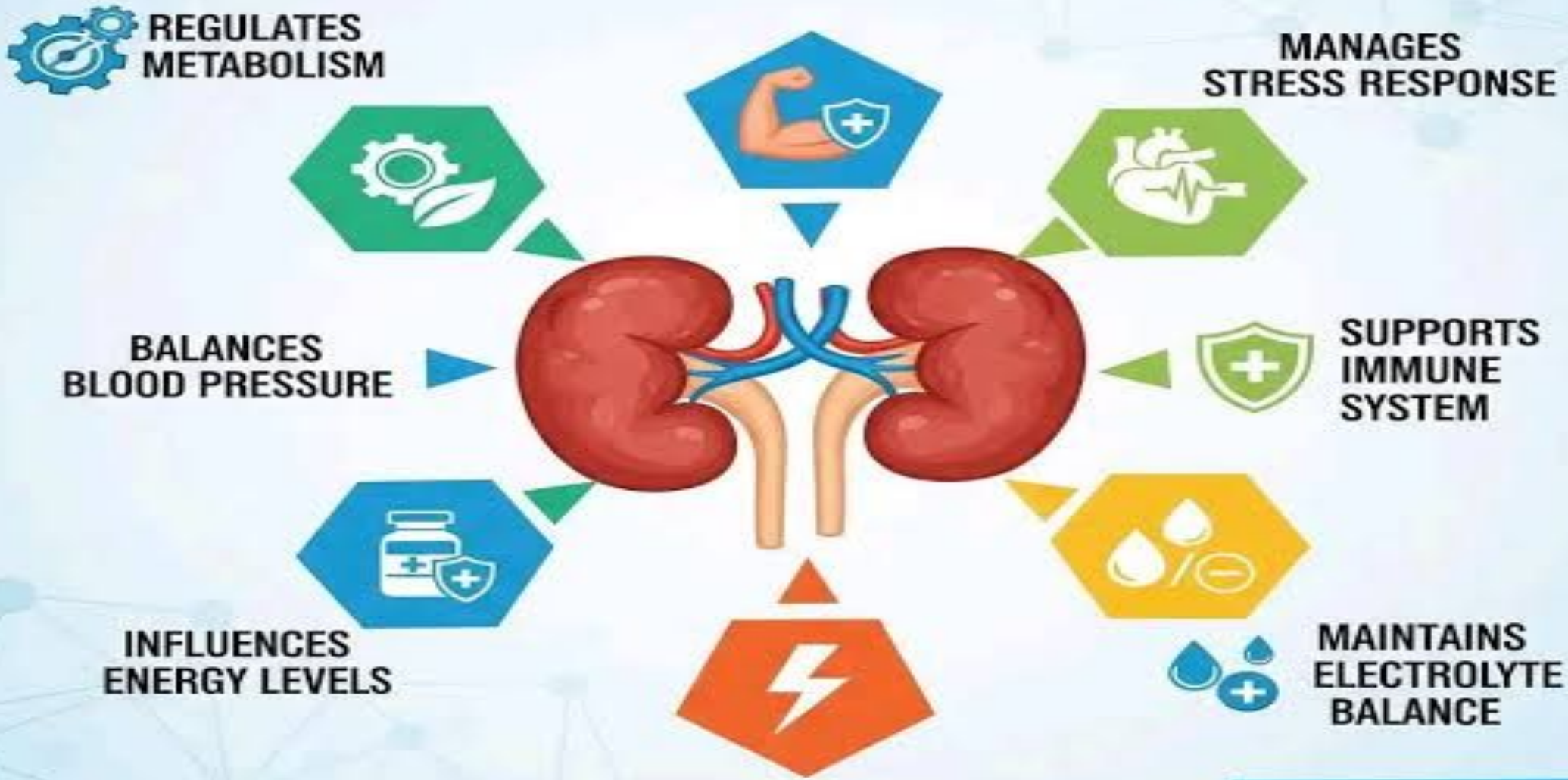
The Steroid hormones, mineralocorticoids, glucocorticoids, and androgens

Adrenal Medulla:-

The Catecholamine (Norepinephrine, Epinephrine)

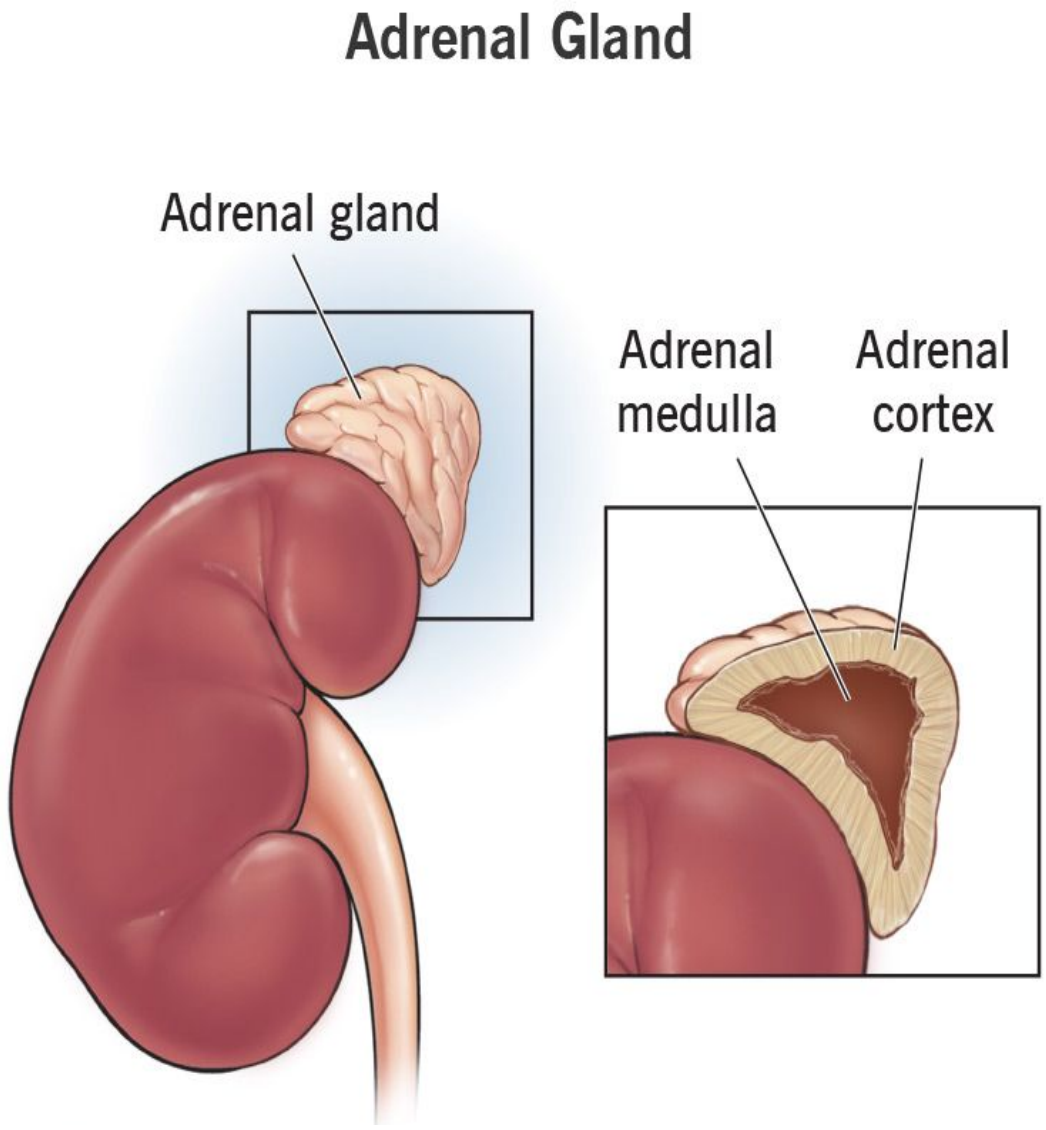
- The AG has 2 different embryologic origins, which influence the mechanisms that control hormone production by each of the 2 components.

BENEFITS OF ADRENAL GLAND



Part -1

Adrenal Cortex



Functional Anatomy and Zonation

The Adrenal Cortex is the outer part of the adrenal gland that makes steroid hormones essential for life.

Three Zones and Functions

The cortex has three distinct layers from the outside in. Each layer makes different hormones.

The adrenal cortex secretes three main categories of steroid hormones, commonly referred to as corticosteroids: mineralocorticoids, glucocorticoids, and androgens (sex hormones).

Zona glomerulosa (outer layer):

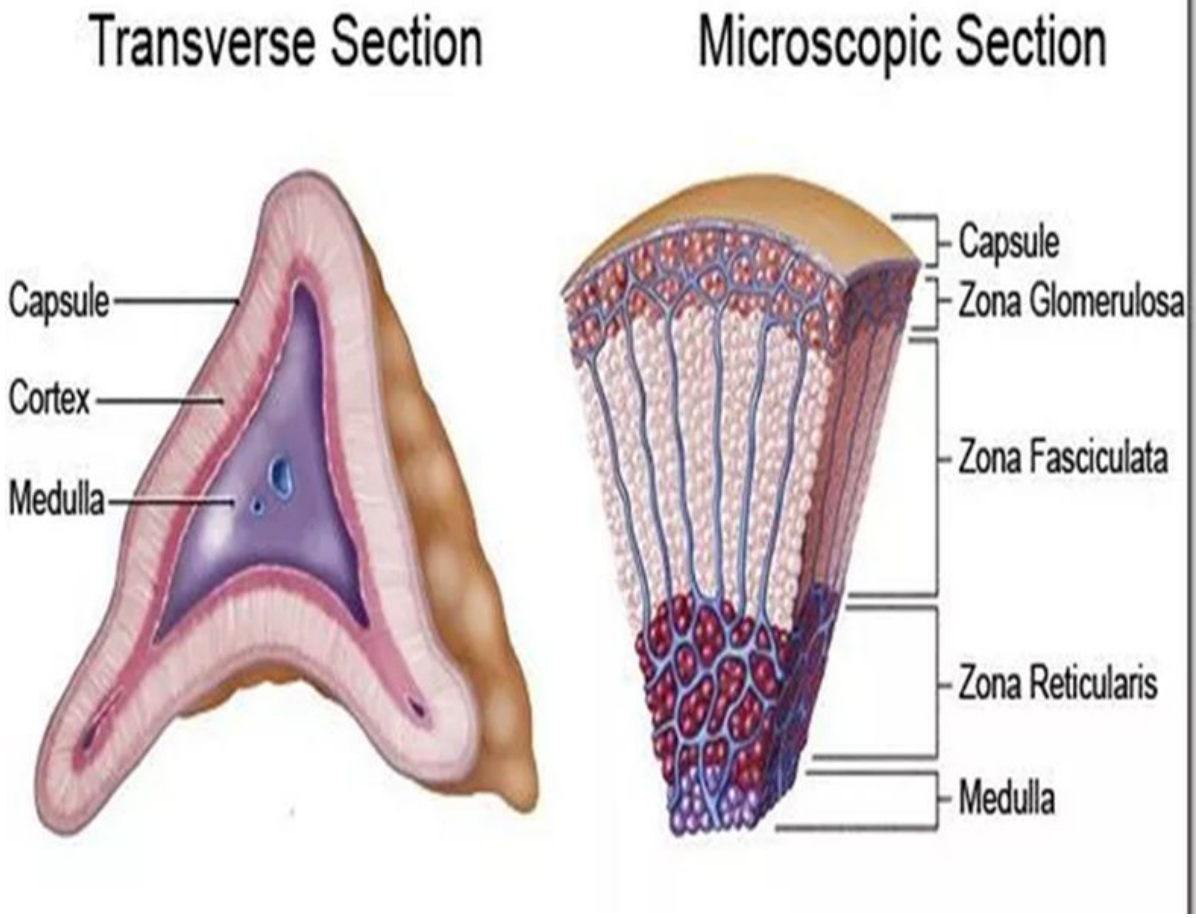
Makes mineralocorticoids like aldosterone.

Zona fasciculata (middle layer):

Makes glucocorticoids like cortisol.

Zona reticularis (inner layer):

Makes sex hormones called androgens.

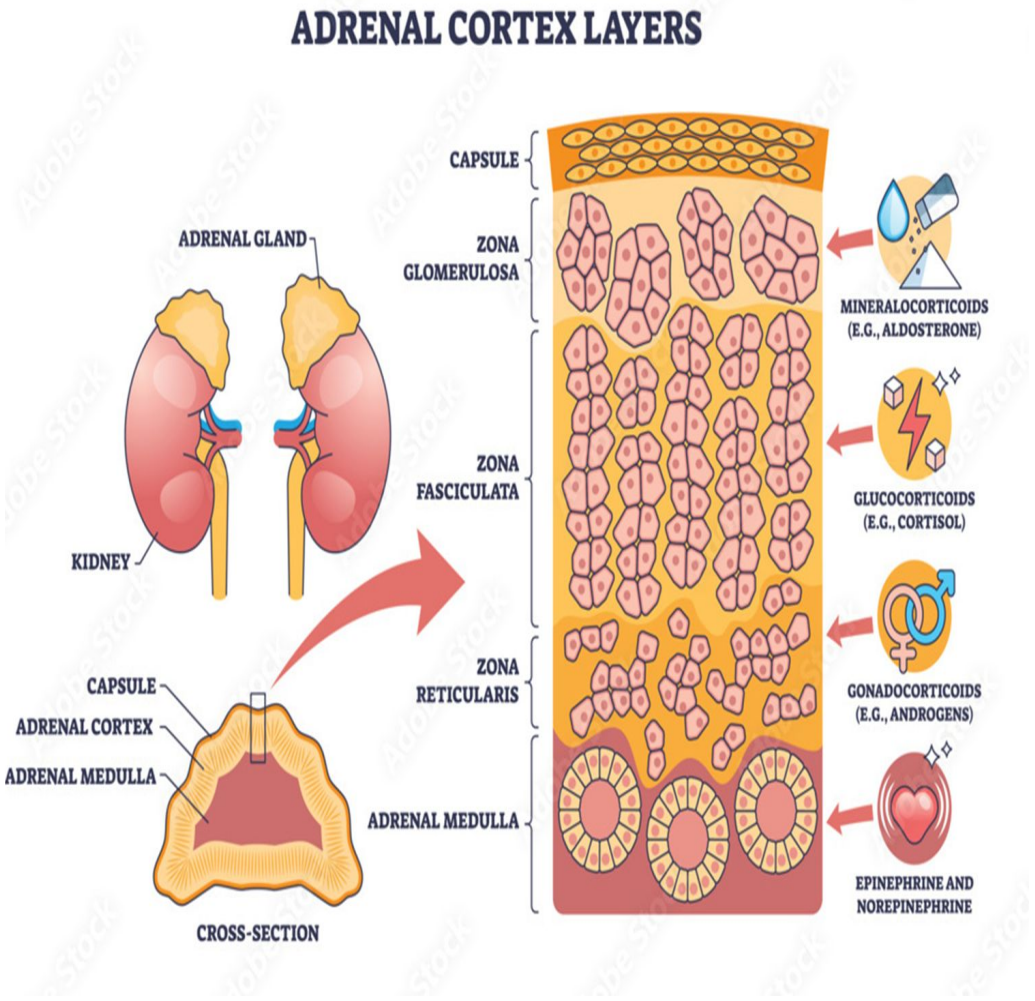


Major Categories of Adrenal Cortex Hormones

- **Mineralocorticoids:**
  - Regulate salt and water balance.
  - Control blood pressure.
  - Main hormone: **Aldosterone** (which tells the kidneys to keep sodium and water).

- **Glucocorticoids:**
  - Help manage the body's use of sugar, fat, and protein.
  - Control the response to physical or mental stress.
  - Lower inflammation and suppress the immune system.
  - Main hormone: **Cortisol**.

- **Androgens:**
  - Act as weak male sex hormones.
  - Support sexual development and reproductive function.
  - Main hormones: **DHEA** (DHEA stands for **dehydroepiandrosterone)** and **androstenedione**.



Mineralocorticoid

Mineralocorticoids are the corticosteroids that act on the minerals (electrolytes), particularly sodium and potassium.

Maintains Na+ balance, by reducing excretion of sodium from the body

It stimulates reabsorption of Na+ by the kidneys and K+ excretion

Main Hormones are:-

- Aldosterone-90 %
- Deoxy-corticosterone
- Corticosterpone
- 9 alpha fluorocortisol (synthetic Aldosterone) Cortisol
- Cortisone

Functions of the Aldosterone

The functions of aldosterone, a hormone primarily produced by the adrenal glands, are crucial for maintaining the body's electrolyte balance, fluid volume, and blood pressure. Here are some key functions of aldosterone:

On Kidneys Sodium Reabsorption

Aldosterone, a mineralocorticoid, causes the Kidneys to conserve Sodium ions and thus water, and to excrete Potassium ions

Aldosterone Increases Renal Tubular Reabsorption of Sodium and Secretion of Potassium by acting on the **distal tubules and collecting ducts of the nephrons** in the kidneys, increasing the activity of sodium channels in these tubules.

High levels of aldosterone decreases Na loss and increase K loss. By retaining sodium, aldosterone helps to regulate extracellular fluid volume and blood pressure.

Potassium Excretion

In addition to promoting sodium reabsorption, aldosterone stimulates the excretion of potassium ions in the kidneys. It does so by increasing the secretion of potassium into the renal tubules, which is then eliminated from the body through urine. This process helps to maintain appropriate levels of potassium in the bloodstream, preventing hyperkalemia (high potassium levels) which can have serious health consequences.

Hydrogen Ion Excretion

Aldosterone also facilitates the excretion of hydrogen ions (H+) in the kidneys. By promoting the secretion of hydrogen ions into the renal tubules, aldosterone indirectly helps to regulate blood pH and maintain acid-base balance within the body.

Functions of Aldosterone

Water Retention

The reabsorption of sodium ions by aldosterone in the kidneys leads to increased osmolarity of the extracellular fluid. This osmotic effect causes water to be retained along with sodium, thereby expanding the volume of extracellular fluid.

Regulation of Blood Pressure

Through its actions on sodium and water retention, aldosterone plays a crucial role in regulating blood pressure.

Aldosterone is secreted in response to decreasing blood volume and blood pressure as a result of changes in the kidney.

Person develops hypertension which persist as long as exposed to high levels of Aldosterone.

Response to Stress and Dehydration

Aldosterone secretion is stimulated by various factors, including decreased blood volume, low blood pressure, and increased potassium levels.

In response to stress, dehydration, or other conditions that threaten fluid and electrolyte balance, aldosterone secretion increases to conserve sodium and water and restore normal blood volume and pressure.

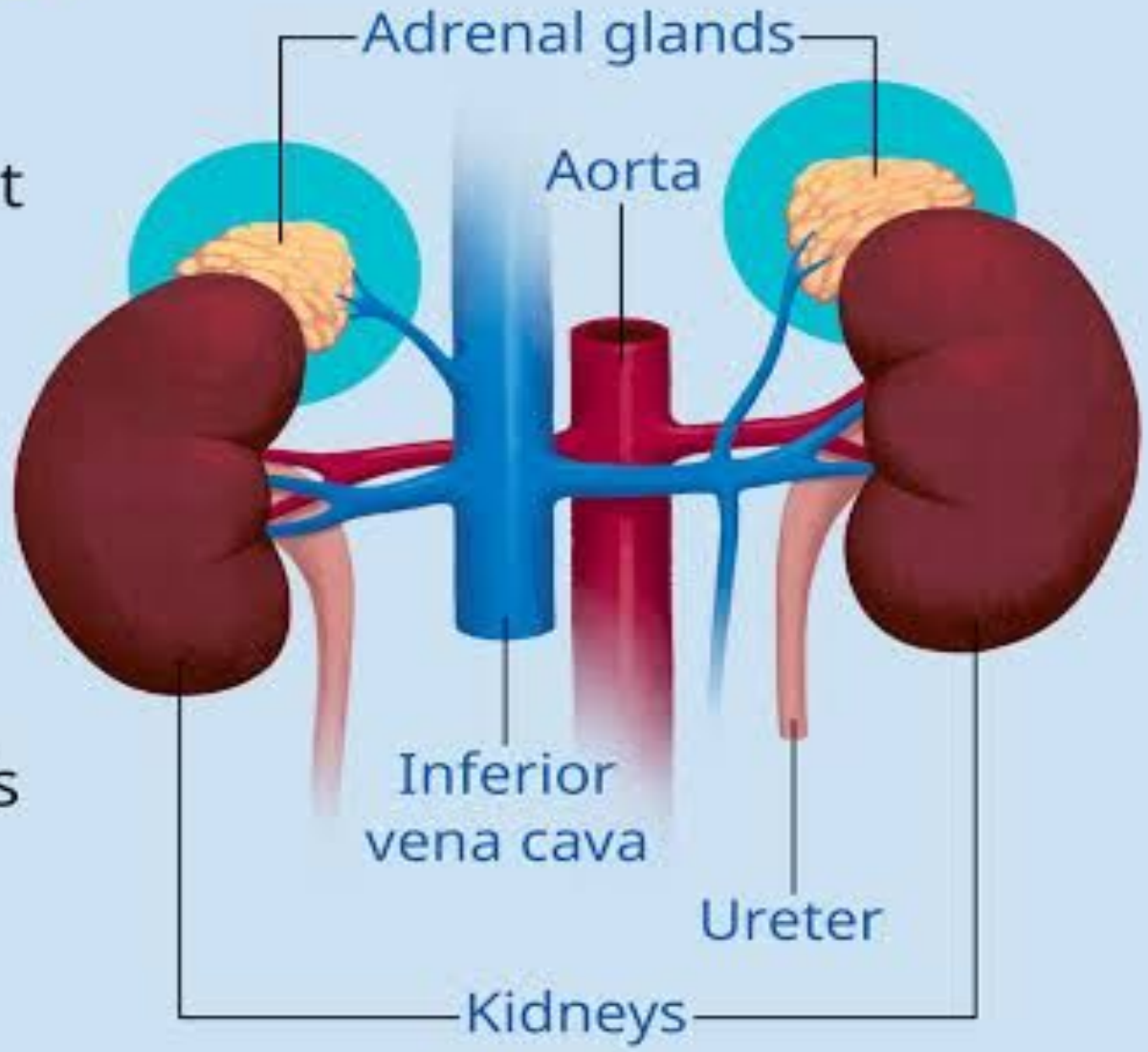
(On sweat and salivary glands)

Aldosterone stimulates Sodium and Potassium transport in Sweat glands, Salivary glands and Intestinal Epithelial cells.

Aldosterone increases the reabsorption of NaCl and secretion of K by ducts to conserve body salts in hot environment.

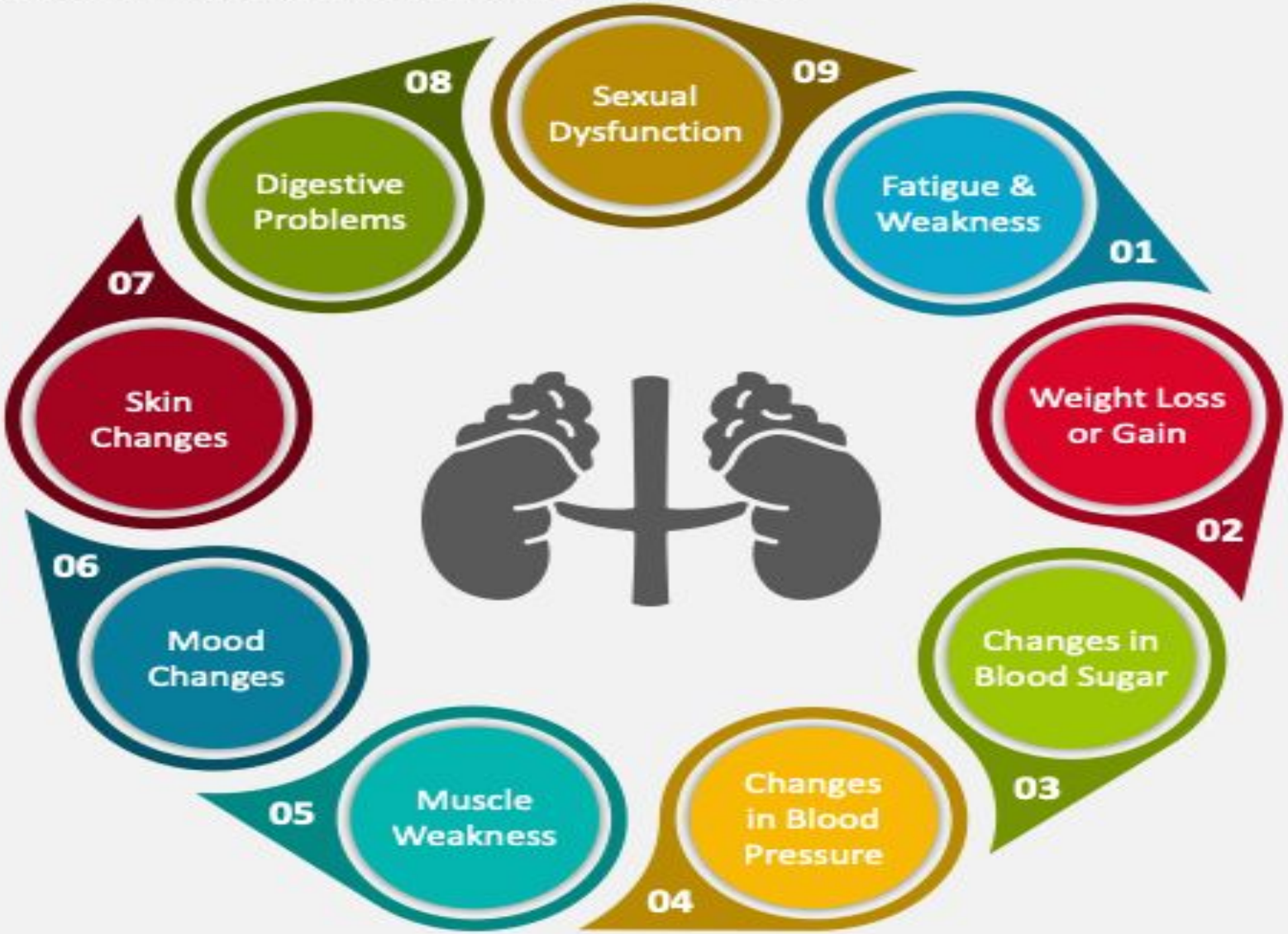
Aldosterone Function

- Stress response
- Childhood sexual development
- Supporting the body while pregnant
- Using nutrients from food
- Balancing salt and water levels in the body
- Managing blood glucose levels
- Manages blood pressure
- Maintains balance between fluids and salt



ADRENAL GLAND

Signs and Symptoms of Adrenal Gland Problems

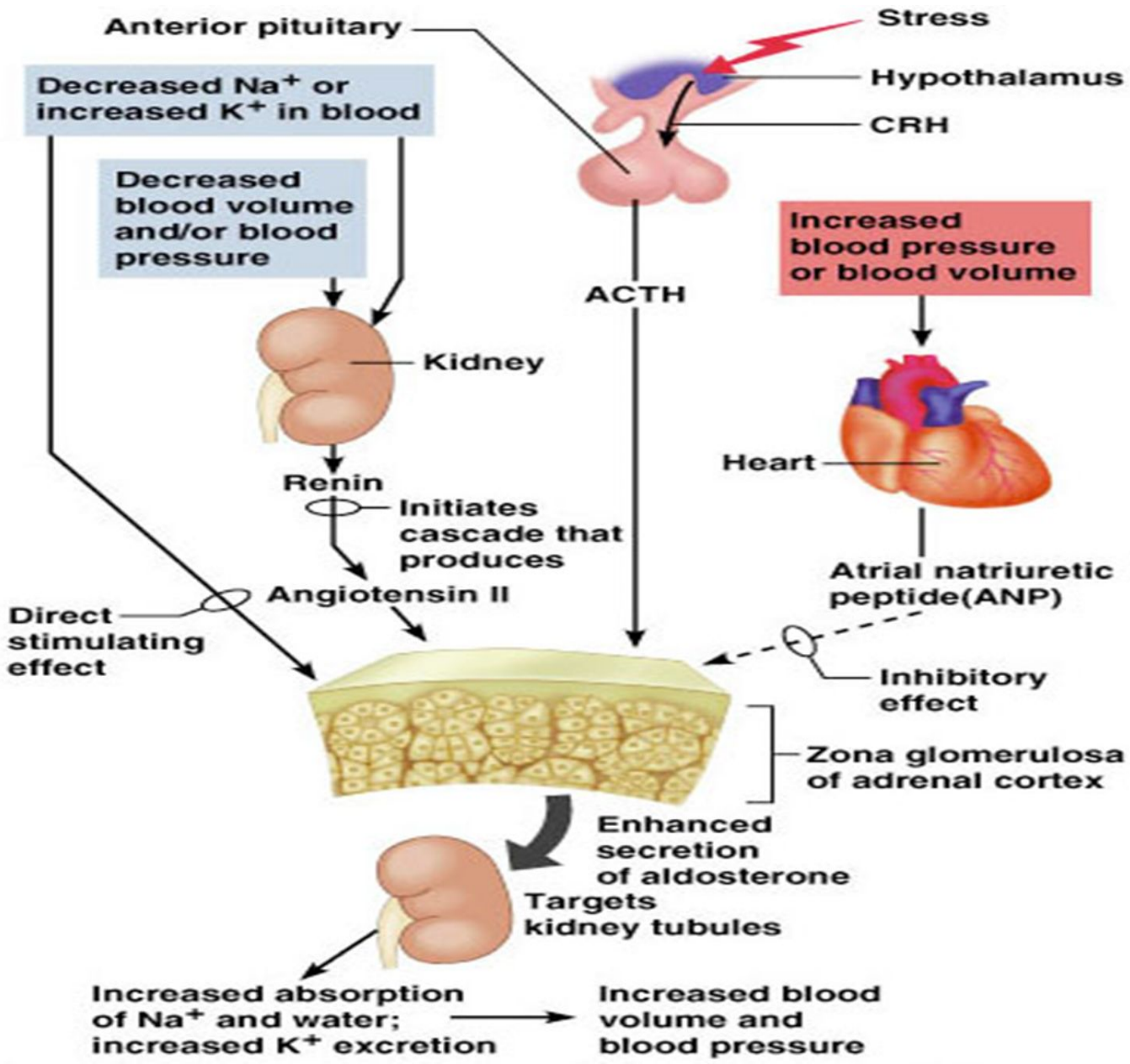


Regulation Of Aldosterone

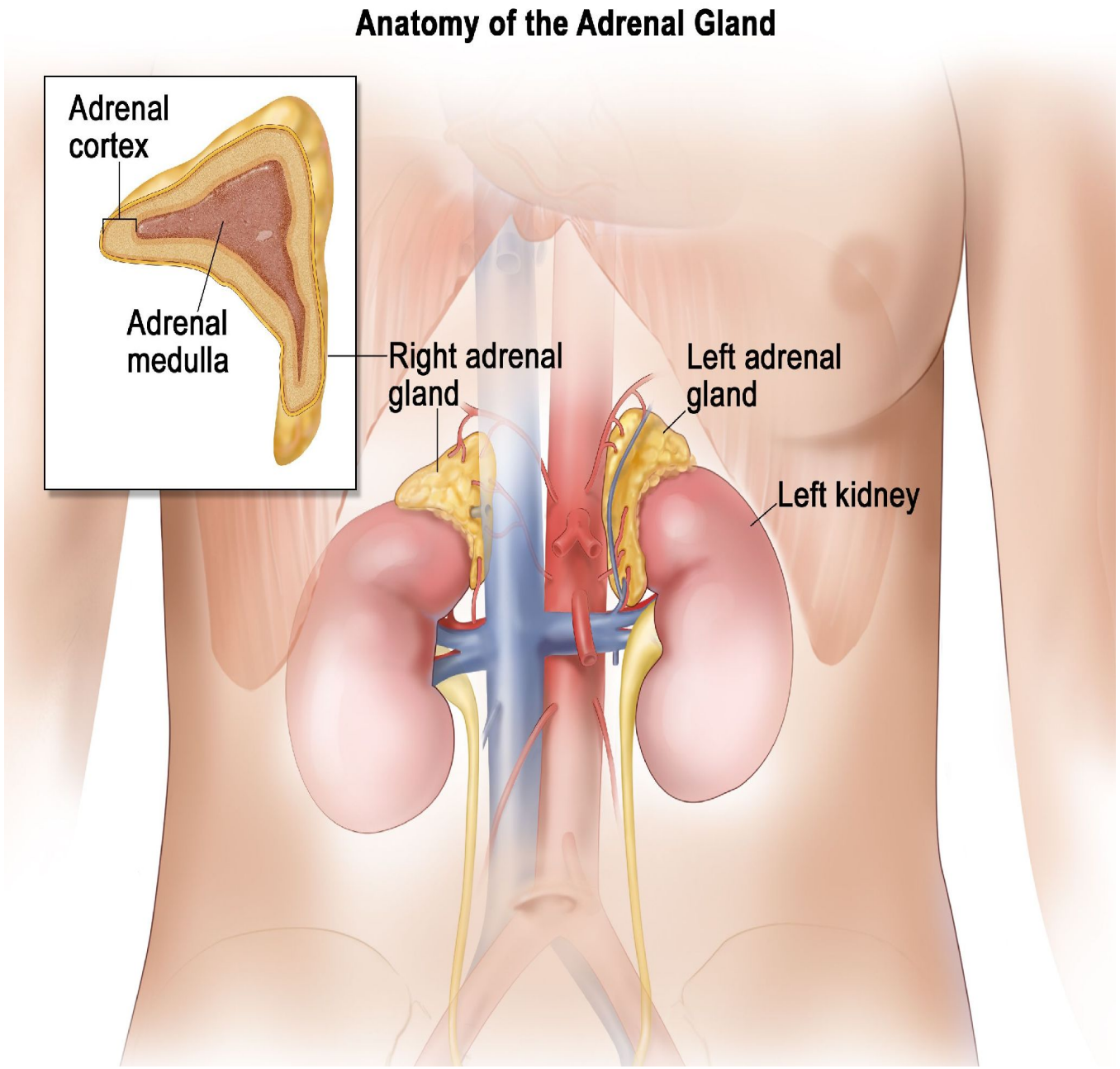
Aldosterone secretion is stimulated by four factors:

- Rising blood levels of K+
- Decreasing blood volume or pressure (renin-angiotensin system).
- Low blood Na+
- ACTH

The Four Mechanisms of Aldosterone Secretion



ADRENAL MEDULLA



ADRENAL MEDULLA

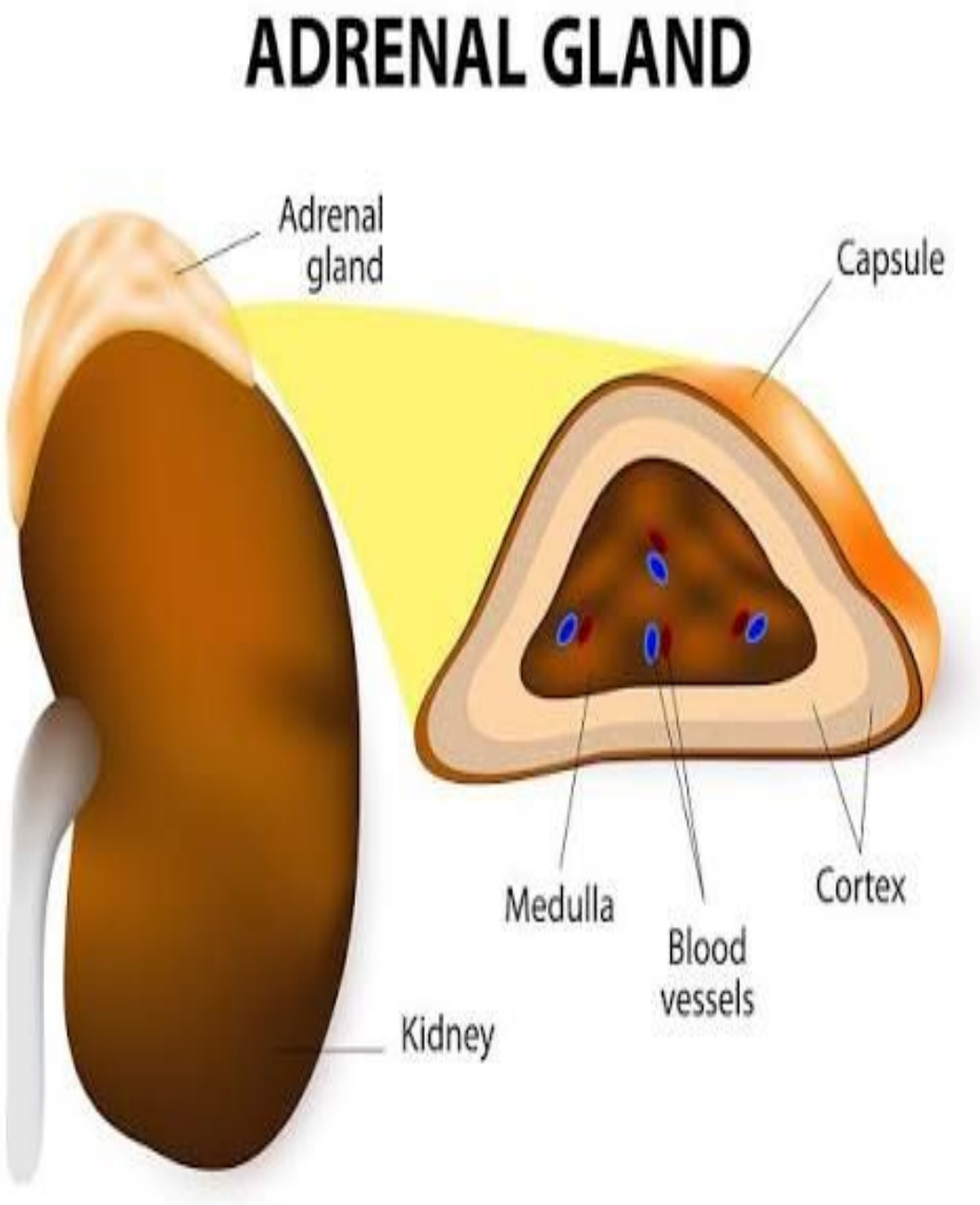
INTRODUCTION

Medulla is the inner part of adrenal gland and it forms 20% of the mass of adrenal gland.

It is made up of interlacing cords of cells known as chromaffin cells. Chromaffin cells are also called pheochrome cells or chromophil cells. These cells contain fine granules which are stained brown by potassium dichromate.

The **adrenal medulla** secretes stress-response hormones into the bloodstream

Position: It sits at the center of each adrenal gland, located just above the kidneys.



Adrenal Medulla

The function of adrenal medulla is governed by nervous system and the chromaffin cells release the hormones responsible for various metabolic reactions.

Release of Adrenaline and Noradrenaline is triggered by nervous stimulation in response to physical or mental stress.

Adrenaline (epinephrine) acts primarily as a stress hormone released by the adrenal glands, whereas

noradrenaline (norepinephrine) functions mainly as a neurotransmitter in the nervous system while also serving as a hormone

Adrenal Medullary Hormones

Adrenal Medullary hormones are involved in control of

- Heart rate,
- Blood pressure,
- Blood vessel constriction in various organs like skin, gastrointestinal tract
- Dilation of blood vessel in skeletal muscles.

Adrenal Medulla

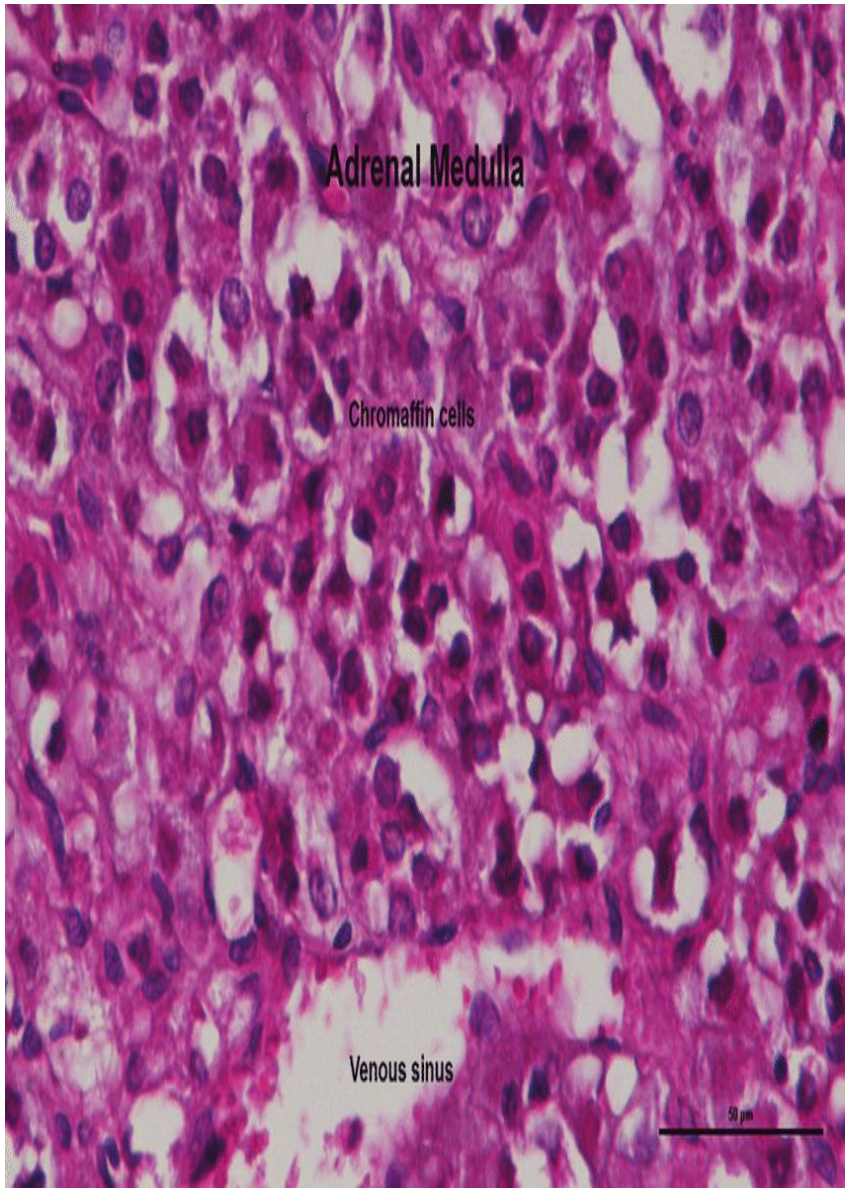
HORMONES OF ADRENAL MEDULLA

Adrenal medullary hormones are the amines derived from catechol and so these hormones are called **catecholamines.**

Catecholamines secreted by adrenal medulla are

1. Adrenaline or epinephrine - 80%
2. Noradrenaline or norepinephrine - 20%
3. Dopamine

Secretion of these hormones causes Blood to be diverted to the Brain ,Heart and skeletal muscles.



Functions of catecholamines

- Glucogenolysis in Liver and skeletal muscles (can lead to hyperglycemia which increases blood glucose level.
- Increase heart rate and blood pressure.
- Causes vasoconstriction of blood vessels
- Metabolization of free fatty acids
- Increase Metabolic rate
- Increase O2 consumption

The three primary catecholamines in the body are:

- **Dopamine:**

Controls movement, memory, mood, and the brain's reward system.

- **Norepinephrine (Noradrenaline):**

Increases heart rate and blood pressure to help the body respond to stress.

- **Epinephrine (Adrenaline):**

Powers the immediate fight-or-flight response by sending extra blood to the heart, lungs, and muscles

Dopamine

- Dopamine is a type of neurotransmitter and hormone. It is secreted from the adrenal medulla as well as from the hypothalamus.
- It plays a role in many important body functions, including, memory and pleasurable reward and motivation mood, learning, sleep, memory, lactation, and involuntary movement.
- High or low levels of dopamine are associated with several mental health and neurological illness.

Epinephrine

Epinephrine (Adrenaline) Main role:

Acts primarily as a hormone released by the adrenal glands.

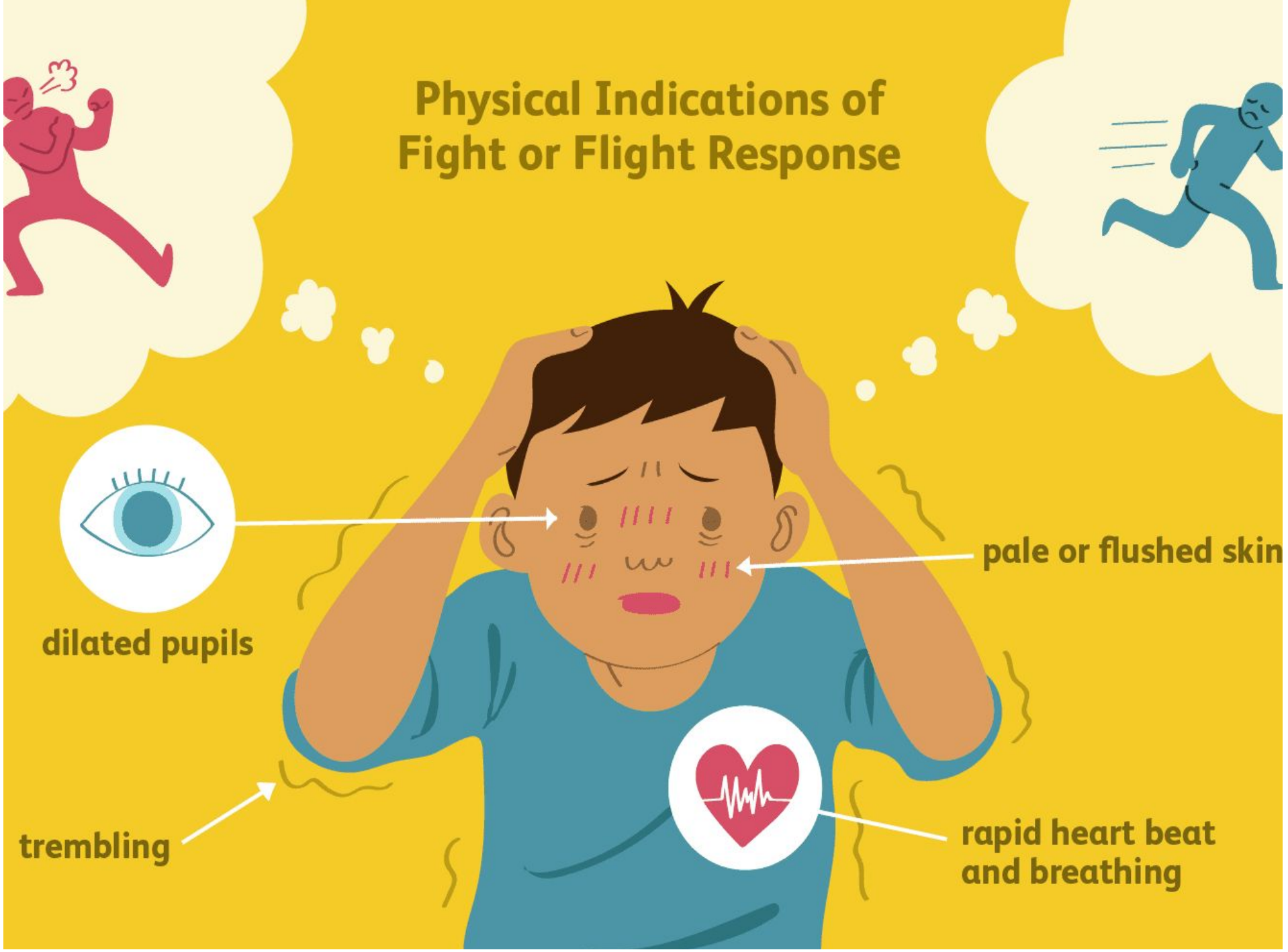
Heart and lungs: Increases heart rate and opens airways to improve breathing.

Metabolism: Breaks down glycogen into glucose to spike blood sugar and energy.

Medical use: Treats severe allergic reactions (anaphylaxis), severe asthma, and cardiac arrest.

Abnormal levels are linked to sleep disorders, anxiety, hypertension and lowered immunity.

Epinephrine's major action is in its role as a hormone. Epinephrine is released by your adrenal glands in response to stress.


Physical Indications of
Fight or Flight Response
dilated pupils
pale or flushed skin
trembling
rapid heart beat
and breathing

Noradrenaline

Norepinephrine (Noradrenaline) Main role:

Acts primarily as a neurotransmitter in the sympathetic nervous system and partly as a hormone

Norepinephrine, also called noradrenaline, is both a neurotransmitter and a hormone.

As a neurotransmitter, it's a chemical messenger that helps transmit nerve signals across nerve endings to another nerve cell, muscle cell or gland cell.

As a hormone, it's released by adrenal glands, which are hat-shaped glands that sit on top of each kidney.

Blood pressure: Constricts blood vessels to maintain stable blood pressure during stress.Increases heart rate and blood pressure to help the body respond to stress.

Brain function: Increases alertness, arousal, attention, and focus.

Medical use: Administered as an emergency injection to treat dangerously low blood pressure and shock

This will boost the supply of oxygen and glucose to the brain and muscles, and increase skeletal muscle readiness.
This prepares your body to react to stressors.

Abnormalities OR Disorders

Disorders of adrenal glands

- **Hyperactivity:-**
- 1. Cushing's syndrome
- 2. Hyperaldosteronism
- 3. Adrenogenital syndrome
- **Hypoactivity:-**
- 1. chronic adrenal insufficiency:Addison's disease
- 2. acute adrenal insufficiency: Addisonian crisis or Adrenal crisis
- 3. Congenital adrenal hyperplasia

Abnormalities of Adrenocortical Secretion

There are many types of adrenal gland disorders, including:

Addison's Disease:

a condition in which the adrenal glands don't make enough cortisol. It is due to the Underproduction of cortisol and aldosterone, often due to an autoimmune response

Cushing's Syndrome:

a condition caused by too much cortisol in the body, often from taking steroid medicines for a long time. It is due to the overproduction of cortisol by the adrenal glands.

Pheochromocytoma:

a rare tumor of the adrenal medulla that produces too much adrenaline and noradrenaline.

Adenoma:

a benign tumor (not cancer) that makes too much aldosterone and may cause serious high blood pressure

Congenital Adrenal Hyperplasia (CAH):

a group of inherited disorders in which the adrenal glands don't make enough cortisol

Hypoadrenalism (Adrenal Insufficiency)

- **Addison's disease** results from an inability of the adrenal cortices to produce sufficient adrenocortical hormones, and this in turn is most frequently caused by primary atrophy or injury of the adrenal cortices.

- Addison's is most commonly caused by autoimmune destruction of the adrenal cortex

- Adrenal gland hypofunction is also frequently caused by tuberculosis destruction of the adrenal glands or invasion of the adrenal cortices by cancer.

- In some cases, adrenal insufficiency is secondary to impaired function of the pituitary gland, which fails to produce sufficient ACTH.

Hypoadrenalism (Adrenal Insufficiency)

Sign and Symptoms

- Melanin Pigmentation
- Hypoglycemia
- Wt. loss,weakness,nausea and vomiting
- Decreased Axillary and Pubic hairs (caused by deficiency of the Androgen hormones)
- ECF volume contraction hypotension,hyperkalemia, and metabolic disorders.



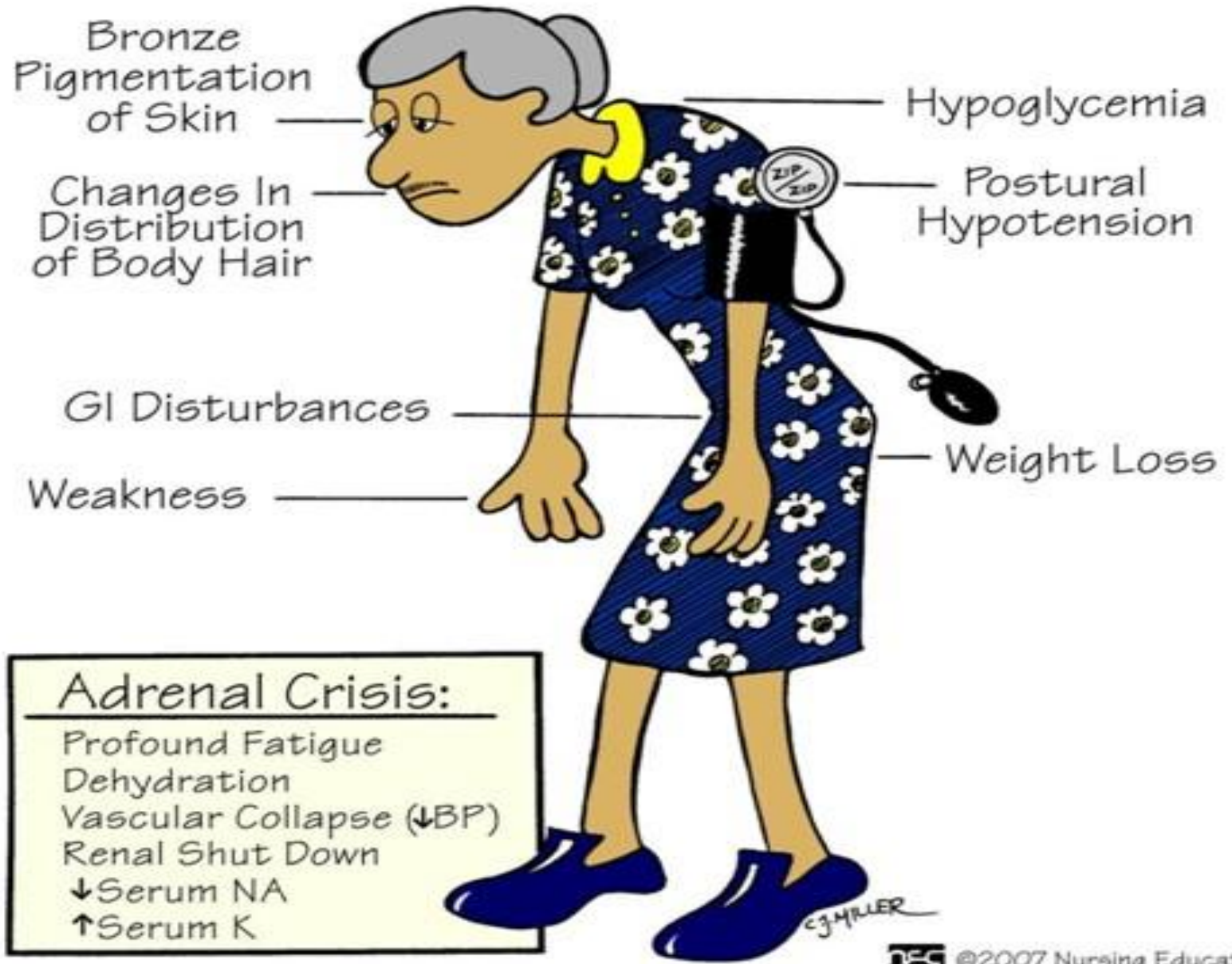






due
in

ADDISON'S DISEASE



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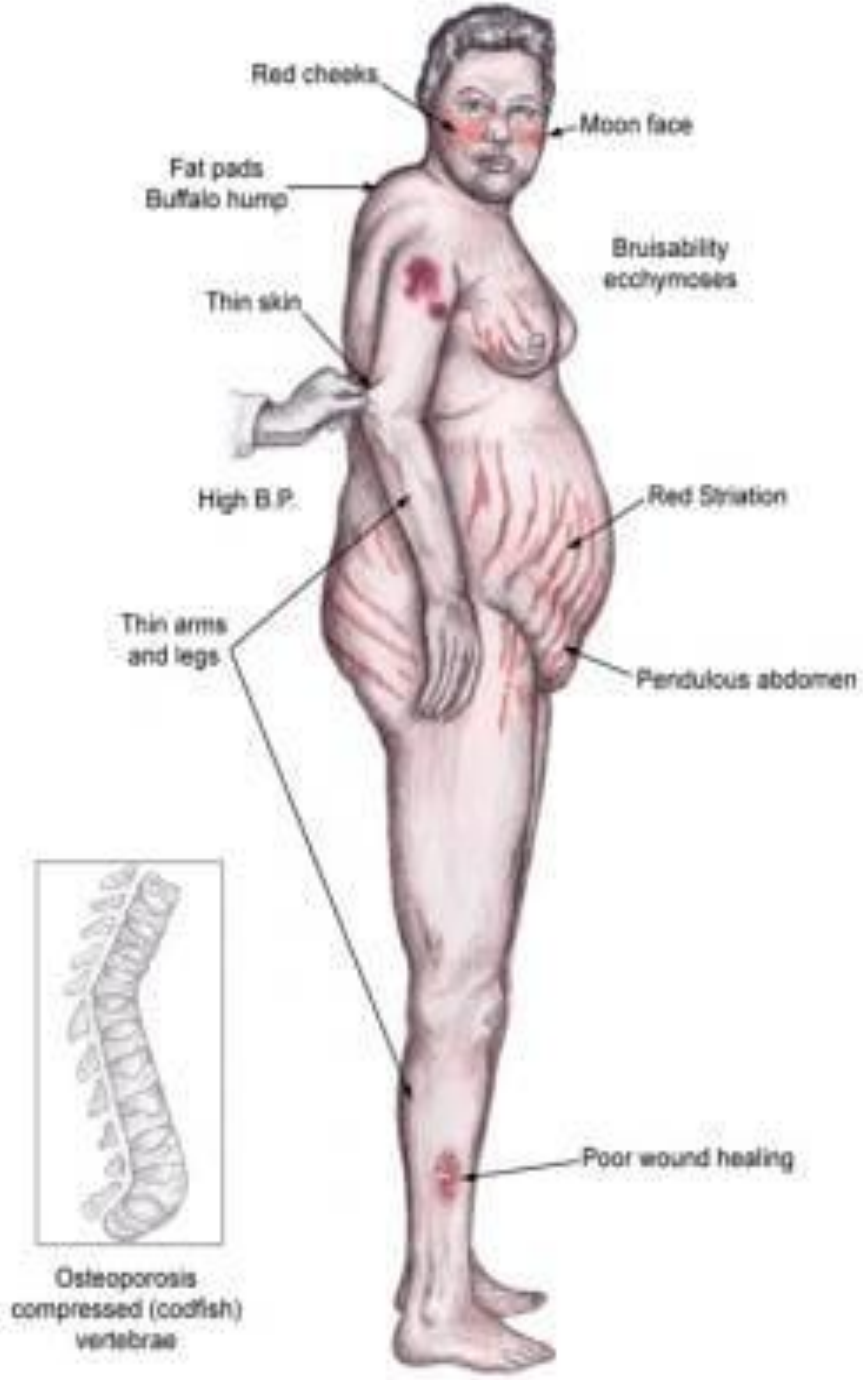
Cushing's Syndrome

Causes:-

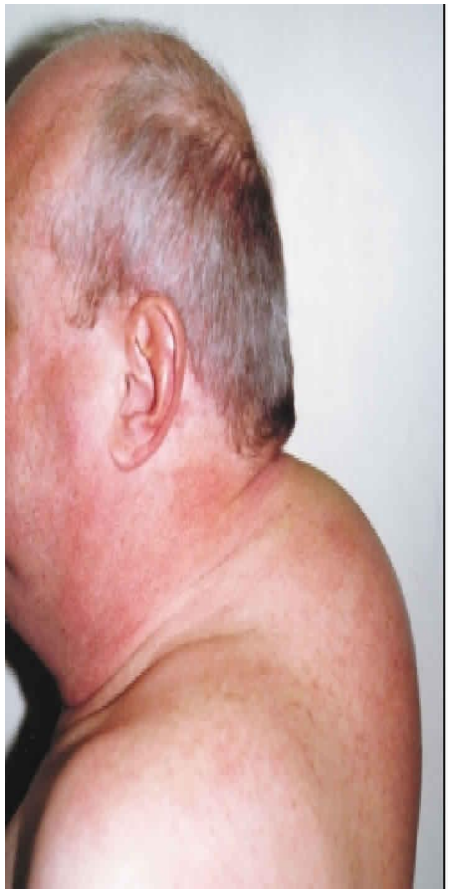
- Increased Plasma Levels of Glucocorticoids (Cortisol) Produce Cushing's Syndrome and is characterized by high plasma levels of ACTH and cortisol
- Cushing's syndrome can also occur when large amounts of glucocorticoids are administered over prolonged periods for therapeutic purposes.

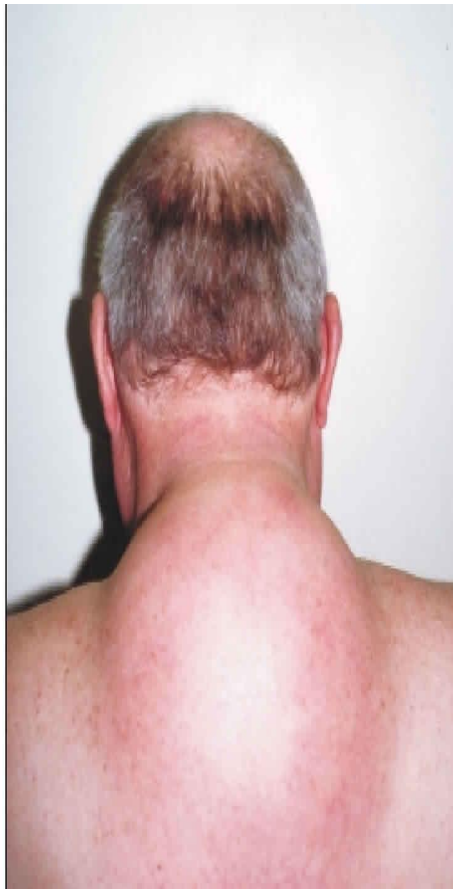
For example, patients with chronic inflammation associated with diseases such as rheumatoid arthritis are often treated with glucocorticoids and may develop some of the clinical symptoms of Cushing's syndrome.

- Mobilization of fat from the lower part of the body, with concomitant extra deposition of fat in the thoracic and upper abdominal regions, giving rise to a buffalo torso or buffalo hump.
- An edematous appearance of the face , acne and hirsutism (“moon face”)
- Hypertension, hyperglycemia, severe muscle weakness, large purplish striae, severe osteoporosis with consequent weakness of the bones



Cushing's syndrome







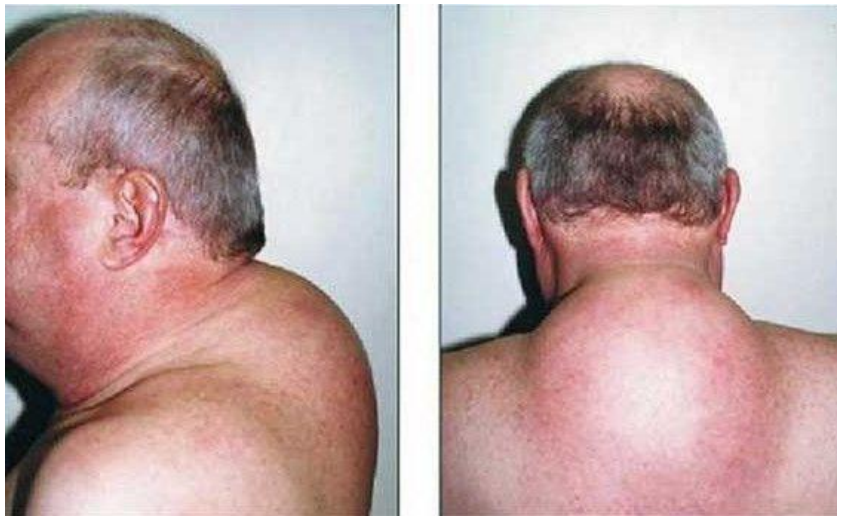
(a) Patient before onset.

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(b) Same patient with Cushing's syndrome. The white arrow shows the characteristic "buffalo hump" of fat on the upper back.

Cushing's syndrome, Cushing's disease









Mineralocorticoid Deficiency

It results in

- Excessive loss of sodium
- Hypovolemia
- Hypotension
- Excessive potassium retention and hyperkalemia
- Mild acidosis

Glucocorticoid Deficiency

- Abnormal carbohydrate, fat, and protein metabolism resulting in muscle weakness, fasting hypoglycemia, and impaired utilization of fats for energy.

- Loss of appetite and weight loss.

- Poor tolerance to stress.